

TAK Server Deployment

Create a new user - Optional

- While ssh'd into your server do the following:
 - Create a new user named atak
 - **sudo useradd -m atak**
 - Set the password for the atak user
 - **sudo passwd atak**
 - Enter the password for the atak user
 - **atakatak**
 - It will ask you to enter it twice
 - If done correctly you should see the following:
 - Make atak user 'sudo user'
 - **sudo usermod -aG wheel atak**
 - Switch from the Root user to the ATAK user
 - **su - atak**

```
[takadmin@localhost ~]$ sudo useradd -m atak
[takadmin@localhost ~]$ sudo passwd atak
Changing password for user atak.
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: all authentication tokens updated successfully.
[takadmin@localhost ~]$
```

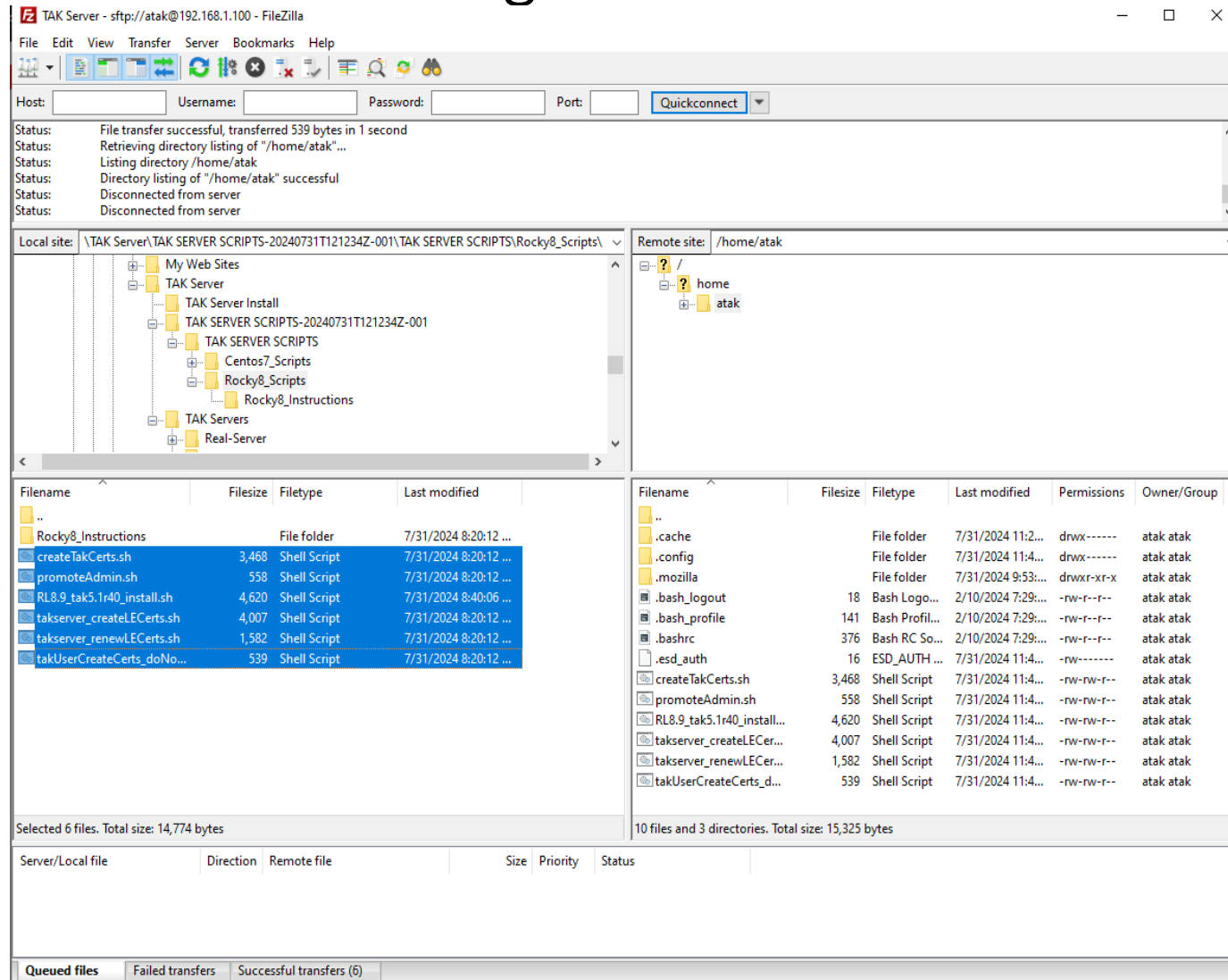
Copy over scripts

- Download FileZilla
 - <https://filezilla-project.org/download.php?platform=win64>
- Once opened:
 - The Local site is your development machines libraries.
 - The Remote site is where the libraries will be for the machine you remote into.
 - In the upper left select this icon:
 - Select “New site”
 - Protocol: SFTP
 - Host: <IP address>
 - User: takadmin
 - Password: atakatak
 - Select “Connect”

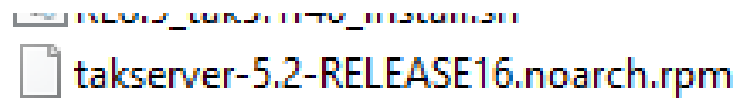


Transferring Scripts

- In the Remote site panel navigate to the following:
 - home → (user)
- In the Local site side navigate to the following:
 - TAK Server Scripts
→ Rocky_Scripts
- From the Local site select the 6 shell scripts and drag and drop them into the (user) folder on the Remote site.



Transferring Install File

- In FileZilla on the Local site panel navigate to the downloaded TAK Server installation file:
- Drag and drop this file into the Remote site  folder.
- You can now minimize the FileZilla application.
- The following should be the contents of your atak folder:

Filename	Filesize	Filetype	Last modified	Permissions	Owner/Group
..					
.cache		File folder	7/31/2024 11:2...	drwx-----	atak atak
.config		File folder	7/31/2024 11:4...	drwx-----	atak atak
.mozilla		File folder	7/31/2024 9:53:...	drwxr-xr-x	atak atak
.bash_logout	18	Bash Logo...	2/10/2024 7:29:...	-rw-r--r--	atak atak
.bash_profile	141	Bash Profil...	2/10/2024 7:29:...	-rw-r--r--	atak atak
.bashrc	376	Bash RC So...	2/10/2024 7:29:...	-rw-r--r--	atak atak
.esd_auth	16	ESD_AUTH ...	7/31/2024 11:4...	-rw-----	atak atak
createTakCerts.sh	3,468	Shell Script	7/31/2024 11:4...	-rw-rw-r--	atak atak
promoteAdmin.sh	558	Shell Script	7/31/2024 11:4...	-rw-rw-r--	atak atak
RL8.9_tak5.1r40_install...	4,620	Shell Script	7/31/2024 11:4...	-rw-rw-r--	atak atak
takserver-5.2-RELEASE...	543,921,018	RPM File	7/31/2024 11:5...	-rw-rw-r--	atak atak
takserver_createLECer...	4,007	Shell Script	7/31/2024 11:4...	-rw-rw-r--	atak atak
takserver_renewLECer...	1,582	Shell Script	7/31/2024 11:4...	-rw-rw-r--	atak atak
takUserCreateCerts_d...	539	Shell Script	7/31/2024 11:4...	-rw-rw-r--	atak atak

Install the TAK Server

- Go to the user folder where the scripts and TAK Server rpm files are.
 - **cd /home/takadmin**
- Queue up the TAK Server install
 - **sudo chmod +x RL9.5_tak5.4r14_install.sh**
- Enter in the password for atak if prompted
 - **atakatak**
- Execute the TAK Server installation
 - **./RL9.5_tak5.4r14_install.sh**
 - If when you try to run this command, it returns: “No such file or directory” run the following and then try the script again:
 - **sed -i -e 's/\r\$//' RL9.5_tak5.4r14_install.sh**
 - Filters and transforms the text

TAK Server Install Prompts

- Atak user password for installing Java 17: atakatak
- Install 1 package: y
- Select java option: 1
- Enter the following with no Spaces (you can enter whatever you want)
 - STATE:
 - CITY:
 - ORGANIZATION:
 - ORGANIZATION UNIT:
- Give a name for the Certificate Authority, has to be at least five characters: (anything you want (TAKSERVER))
- Do you want me to move files around: y
- Following this:
 - the server will restart for 90 seconds before it configures the intermediate-ca for use
 - The server will restart for 270 seconds to ensure the TAK Server is fully restarted prior to promoting the admin.pem certificate to the admin role.

TAK Server Configuration/Monitoring

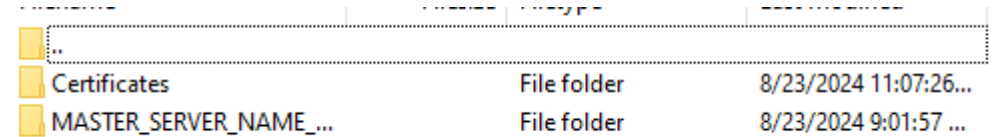
- To enable the server to automatically start when we restart the VM run the following:
 - **sudo systemctl enable takserver.service**
- To start the server from the command line:
 - **sudo systemctl start takserver.service**
- To make sure the server is running:
 - **ls -l /opt/tak/logs/**
 - In here we want to make sure the takserver-messaging.log file is created.
 - **tail -f /opt/tak/logs/takserver-messaging.log**
 - Streams the end of the log to allow us to verify the server is running properly
 - Main things we are looking for:
 - INFO c.b.marti.nio.netty.NioNettyBuilder - Successfully Started Netty Server for TlsServerInitializer on Port 8089 with watermark WriteBufferWaterMark(low: 2048, high: 4096)
 - This lets us know that the client-to-server listener is started and is accepting clients
 - INFO com.bbn.marti.nio.server.NioServer - Server started
 - This tells us the services have started
 - INFO com.bbn.marti.util.VersionBean - TAK Server version 5.0-RELEASE-29-HEAD
 - This lets me know what version is running and that all services have started

Initial Configuration

- If using a different username than atak:
 - Change the directory to the Linux home directory
 - `cd ~`
 - Copy the admin cert to our home directory
 - `sudo cp -v /opt/tak/certs/files/admin.p12 .`
 - Copy the truststore cert to our home directory
 - `sudo cp -v /opt/tak/certs/files/truststore-intermediate-ca.p12 .`
 - Change the ownership to our Linux user
 - `sudo chown -R <username>:<usergroup> /home/<username>`
 - Example: `sudo chown -R takadmin:takadmin /home/takadmin`
 - If you don't do this, you can't copy the files from the server to your development machine.
 - You can use "ls" to verify that the file is in the right place .

Retrieve Admin Certificate

- Open FireZilla
 - Local site: Downloads
 - Remote Site: /home/takadmin
- Locate the “admin.p12” file and the file “trustore-intermediate-ca.p12” in the Remote site panel
 - If you’d like to change the name of the file you can
- Drag and drop the files into the Local site panel Downloads folder.
 - You can create a folder called “Certificates” to keep track of the user certificates.
 - If the permissions are denied:
 - Enter the root user: **sudo su root**
 - Navigate to the file location: **cd /opt/tak/certs/files**
 - Changed permissions: **chmod ugo+rw <filename>**



..		
Certificates	File folder	8/23/2024 11:07:26...
MASTER_SERVER_NAME_...	File folder	8/23/2024 9:01:57 ...

Importing User Certificate

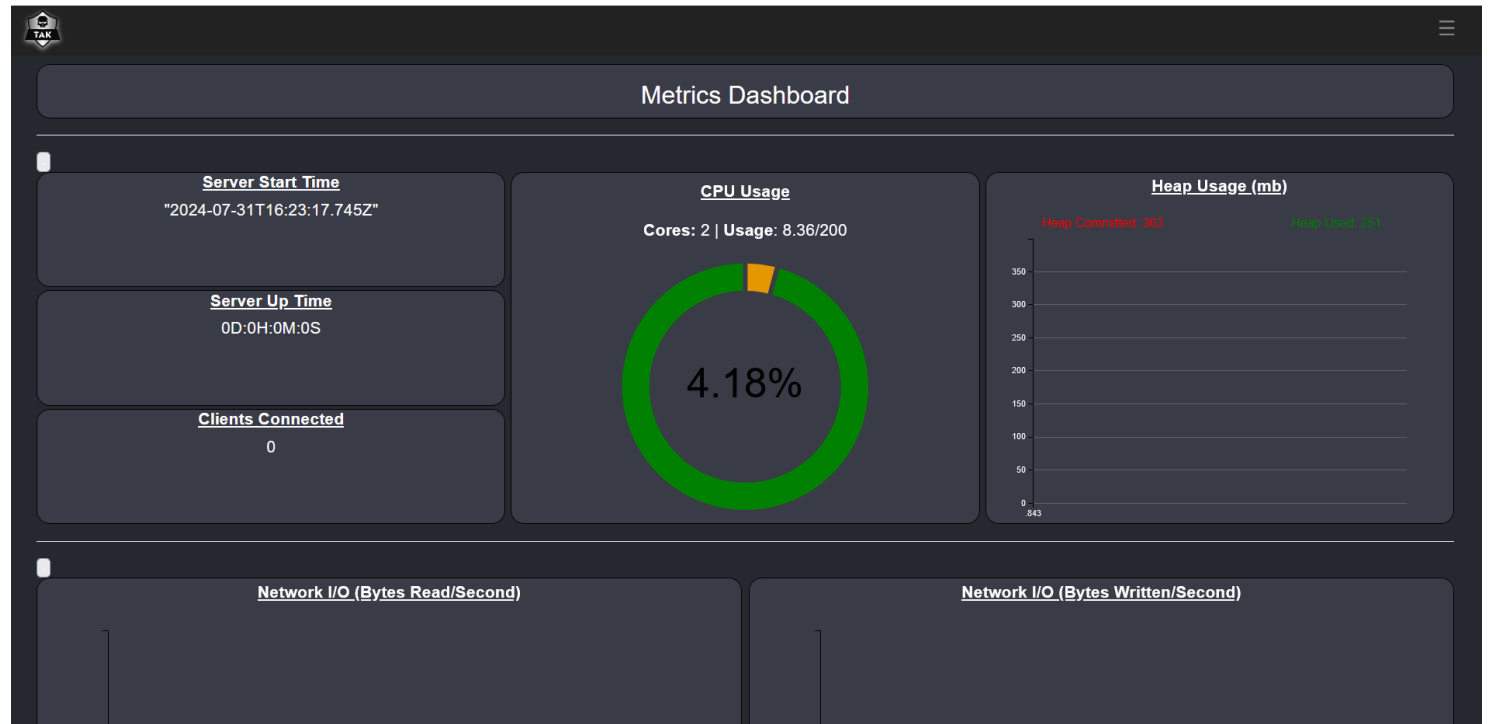
- Windows:
 - Double click on the user certificate, which will open the windows certificate import wizard.
 - Store Location: Current User
 - File path should be correct
 - Password: atakatak
 - Certificate Store: Automatically select the certificate store based on the type of certificate
 - Select “Finish”
 - Select “Yes” for the security warning
 - Import should be successful

Fire Fox Certificate Import

- Open Fire Fox and navigate to its settings
- Search for “cert”
- Select “View Certificates”
- Ensure you are on the “Your Certificates” panel
- Select “Import”
- Navigate to where we copied over the “admin.p12” file and select it.
 - You will be prompted for the password: atakatak
- Once imported you can select “OK” to close the certificate manager.

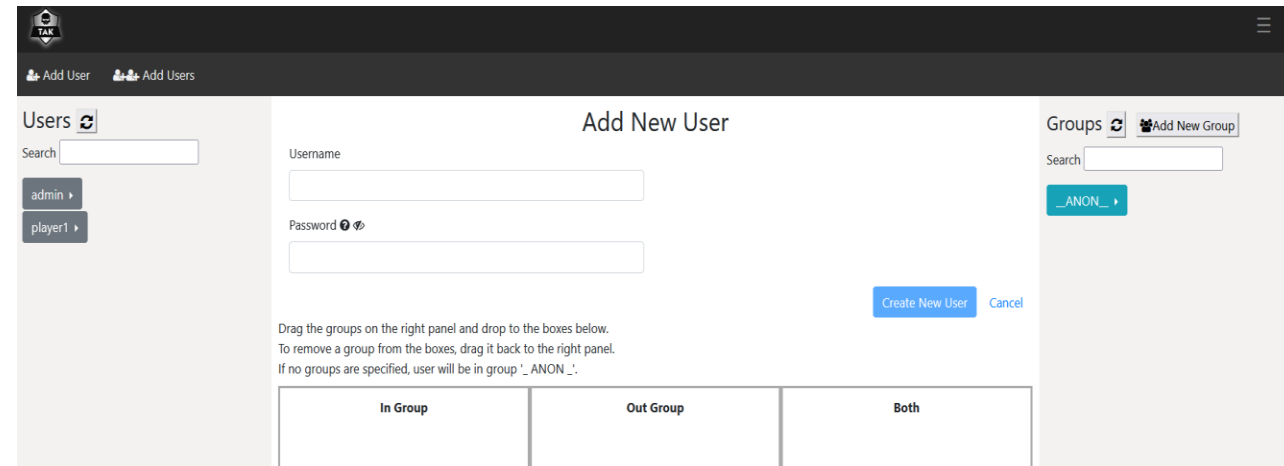
Connect to your TAK Server

- To connect to your TAK Server dashboard, open a your browser.
- Type the following:
 - `https://<IP address>:8443`
- Follow the prompts.
- You are now at your TAK Server dashboard.



Connect to your TAK Server – WebTAK

- To connect to the server as a user:
 - From the server dashboard select the hamburger menu at the top right.
 - Select “Administrative”
 - Select “Manager Users”
 - At the top left select “Add User”
 - Username: <enter username here>
 - Password: <enter password here>
 - Select “Create New User”
 - Open a browser
 - Type: https://<IP address>:8446
 - Sign in as the user you just created



The screenshot shows the 'Add New User' form in the WebTAK interface. The form is divided into three main sections: 'Users' on the left, the main 'Add New User' form in the center, and 'Groups' on the right. The 'Users' section has a search bar and two buttons: 'admin' and 'player1'. The main form has fields for 'Username' and 'Password' (with an eye icon for toggling visibility). Below these fields are three columns: 'In Group', 'Out Group', and 'Both'. The 'Groups' section has a search bar, a dropdown menu showing '_ANON_', and a button to 'Add New Group'. At the bottom right of the main form are buttons for 'Create New User' and 'Cancel'. Instructions at the bottom of the main form state: 'Drag the groups on the right panel and drop to the boxes below. To remove a group from the boxes, drag it back to the right panel. If no groups are specified, user will be in group '_ANON_'.'

In Group	Out Group	Both

Methods of Connecting to TAK Server

- **Option 1 - Create Data Package for Connection**
 - The easiest way to get the client onto the server is to create a Data Package (a zipped file) with the truststore-intermediate-ca.p12 file, a manifest and a preference file with the web address.
 - When you downloaded the MASTER_SERVER_NAME_FOLDER_STRUCTURE it came with a boiler plate profile preference file and a manifest you can modify to reflect this information.
 - Once the data package is created you can share that out with the user's unique username and password you created using the TAK-Server GUI.
 - The end user will download the data package onto their device, use the import feature in the TAK device, upon import the TAK device will prompt for the username and password.
- **Option 2 - Manual cert enrolment**
 - The second option is the TAK client can add a connection manually and enroll for a certificate.
 - To do that share the web address of the server i.e. tak1.mytakserver.com and the username and password you created on the TAK Server web GUI.
 - From the TAK device the user can add the TAK Server connection using the domain address and then choose enroll for certificate and then at prompt enter the username and password combo you issued to them.

Creating Data Packages

- Open file explorer to Master_SERVER_... → Auto Enrollment Data Package
- Copy the “truststore-immediate-ca.p12” file over to the Auto Enrollment Data Package folder
- In the Auto Enroll Data Package folder open the “config” file with a text editor
 - Where it says: “TAKSERVER_NAME_GOES_HERE” remove this and put what you want your server to say (TAK-Server)
 - Where it says: “ipaddress_or_FQDN_goes_here” remove this and put your server IP address. Do not remove :8089:ssl
- Save the document and close it

Creating Data Packages

- In the Auto Enroll Data Package open the MANIFEST folder
- Open the manifest file with a text editor
 - Where it says “Renameit_Server” replace it with what you want to name your data package (TAK-Server)
- Save the file and close the text editor
- We now want to zip the contents in the Auto Enroll Data Package folder.
 - Select all the files (hold ctrl + click all the files desired)
 - Right click
 - Select “Send to”
 - Select “Compressed (zipped) folder”
- If the zipped folder does not match the name set in the manifest file, select the zipped folder and rename it.

Connecting to your TAK Server – Data Package ATAK

- After creating the data package send it to the device you want to import it on.
 - Via: Email, Google Drive, OneDrive, etc.
- Download the zip file onto that device.
- ATAK:
 - Open ATAK.
 - Select the hamburger menu in the top right of the screen.
 - Scroll down and select import.
 - Select Local SD.
 - Select Date to sort the files by date modified.
 - Select the data package zip file you downloaded.
 - Select OK
 - Enter credentials for the server
 - Note the credentials are that which you created on the server already.
 - If all done correctly you should see a message saying you connected successfully

Connecting to your TAK Server – Data Package WinTAK

- If you are not intending to use your current device to run WinTAK
 - After creating the data package send it to the device you want to import it on.
 - Via: Email, Google Drive, OneDrive, etc.
 - Download the zip file onto that device.
- Open WinTAK
 - You can drag and drop the zip file into WinTAK.
 - It will then ask you for the server credentials.
 - Enter credentials for the server
 - Note the credentials are that which you created on the server already.
 - If done correctly it should tell you it has connected.
 - You can always use the import manager accessible from the WinTAK hamburger menu in the top left of the application.

TAK Network Status 

Server login issues

- If you run into the issue where you can't login to the admin portal and are being forced to the webtak login page at port 8446 run the commands below.
 - **cd /home/atak**
 - **sudo chmod +x promoteAdmin.sh**
 - **atakatak**
 - **sudo ./promoteAdmin.sh**

```
promoting admin.pem to administrator
User Updated:
  Username:      'admin'
  Role:          ROLE_ADMIN
  Fingerprint:   2C:16:21:A5:FC:24:99:21:59:4F:E5:49:96:0E:66:CC:4A:FC:C1:
3F:16:4C:C3:9E:D8:2F:D3:57:E2:C5:3C:FE
  Groups (read and write permission):
    __ANON__

restarting tak server
copying admin.p12 to /home/atak/
changing owner of /home/atak/admin.p12 to atak user
+++++
+++++ COMPLETE +++++
+++++
[atak@localhost ~]$
```